

WASP-11b

R-0902

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_171211 · created 2026-07-29T03:51:51+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458098.7543 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.052
Transit depth (Rp/Rs)^2	3.09 +/- 1.81 [%]
Orbital Inclination (inc)	89.74 +/- 1.69
Airmass coefficient 1 (a1)	1.65 +/- 0.97
Airmass coefficient 2 (a2)	-0.46 +/- 0.7
Scatter in the residuals of the lightcurve fit is	58.37 %
Best Comparison Star	#2 - [302, 370]
Optimal Aperture	2.08
Optimal Annulus	11.48
Transit Duration (day)	0.105 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.052 vs 0.131 (deviation 0.87σ)
Duration fit vs published	0.105 d vs 0.1075 d (deviation 0.21σ)
Mid-time O–C	4.5 min
Depth SNR	3.4
Residual scatter	58.37 %

Light curve

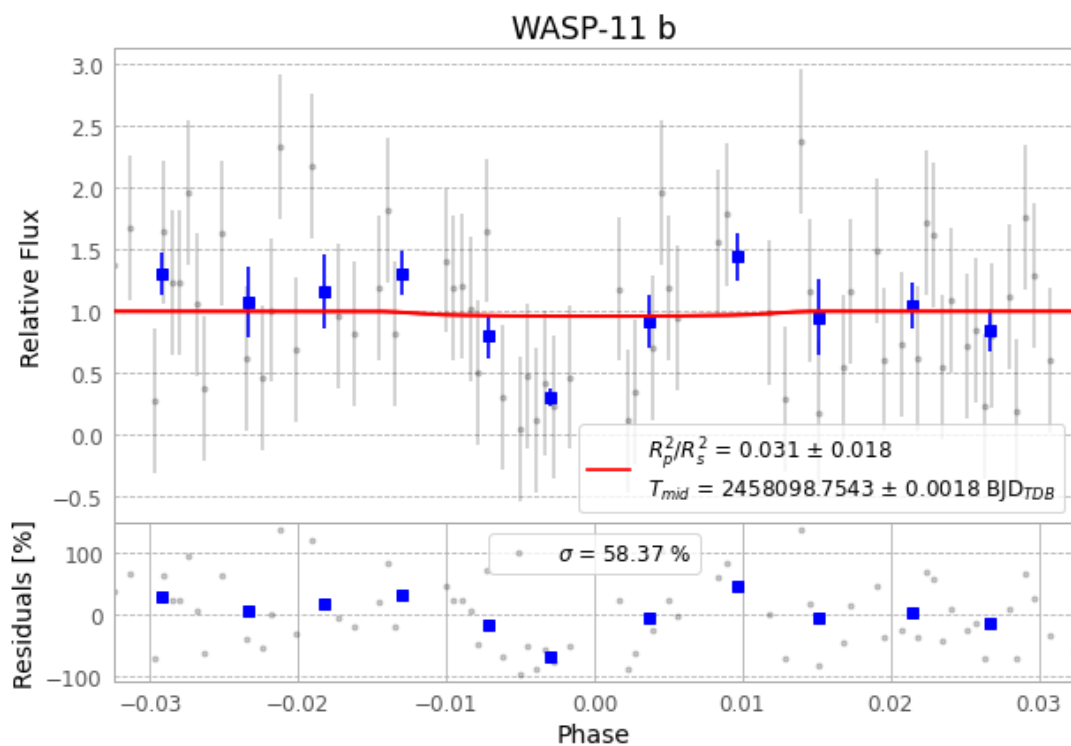


Figure 1 — FinalLightCurve_WASP-11 b_2017-12-10.png (SHA-256 9d953499998f8e48...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 333], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 32af473bf965fe18...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

117 FITS frames (77 MB), HATP-10171211030312.FITS → HATP-10171211085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-171211.log (88fe5436bf1d...), FinalLightCurve_WASP-11 b_2017-12-10.png (9d953499998f...), FinalLightCurve_WASP-11 b_2017-12-10.csv (6aab9d85a4e5...)

Compute resources

Wall time 175.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 03:51Z.